

Tree Ring Climate Detective

Use an authored ring-card code to build and defend a model history, then explain what real evidence would require.

Win condition: Most accurate use of the practice-card code, supported with claim-evidence-reasoning.

THE SCIENCE YOU ARE USING

Rings as proxy data. One annual ring contains earlywood and latewood. Ring width can respond to climate, but it also depends on species, site, age, competition, damage, and other conditions. These six authored cards use stylized widths and black marks as a practice code only. Real dendrochronology cross-dates many trees and calibrates measurements against local weather records.

Claim, evidence, reasoning. Under the printed practice-card code, assign model events to width and mark patterns, then cite the exact annual bands that support each claim. Alternating gray and white fill only separates years; it does not label good or stressful conditions. A single authored card cannot reconstruct real climate.

HOW TO RUN IT

- 01 Learn the ring code.** For these authored cards only, relatively wide means model-favorable, relatively narrow means model-stress, and a black mark means model-disturbance. Do not treat that code as a universal rule for real trees. Each annual band includes earlywood and latewood; gray and white shading only separates adjacent years.
- 02 Read the sequence.** Read the authored annual bands from center to bark and record each width or black-mark pattern in order.
- 03 Infer the events.** Apply the printed card code to assign model events such as stress, a marked disturbance, crowding, or recovery. Label them as model events, not a real tree history.
- 04 Cite your evidence.** For each model inference, point to the exact annual bands that support it on a claim-evidence card.
- 05 Synthesize the story.** Combine the model inferences into one practice-card history, then name the cross-dating and local calibration real climate reconstruction would require.

Safety: Sand wood cookies smooth if used. No cutting tools with students. Allergy: Wood dust sensitivity note if using real samples.

MATERIALS

Printed ring cards	one set per team (4 teams, plus 1 spare)
Answer boards	one per team (4, plus 1 spare)
Wood cookies (optional)	one per team if used (sanded), printed cards are the primary medium
Timers	one per team (4) or the projected room timer
Claim-evidence cards	a stack per team (about 5 to 7 each)

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (which authored pattern is the clearest model-stress code, and why):

Our model history under the practice-card code, oldest band to newest:

The model events we assigned under the printed code:

The exact annual bands that support each model event:

DEFEND YOUR READING

Which ring pattern was easiest to misread, and how did you resolve it?

If you read it again, what is the ONE inference you would re-check, and with which rings?

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Practice-code inferences	40
Evidence justification	20
Speed	20
Final synthesis	20
Total	100